

CHAPTER-1

INTODUCTION

ORIGIN OF FINANCE:

Finance originated around 3000 BCE in ancient Mesopotamia, where temples and places served as early banks for storing valuables, leading to the development of credit, loans, by 1800 BCE formal debt regulation in the code of Hammurabi. It evolved from barter systems to metallic money and banking, formalizing in medieval Italy and establishing modern theory by the 1950s.

1. Ancient beginnings

- Finance began with the barter system, where goods were exchanged directly.
- Around 3000 BCE in Mesopotamia, people started recording transactions on clay tablets.
- Temples and palaces acted like early financial institutions, storing wealth and lending grain or silver.

2. Introduction of money

- The first metal coins were introduced in Lydia around the 7th century BCE
- Money replaced barter, making trade easier and more efficient.
- This led to the growth of markets and economic activity.

3. Birth of banking

- Banking systems emerged in ancient Greece and Rome.
- Modern banking developed in medieval Italy, especially in cities like Florence.
- The Medici family played a major role in developing banking practices like loans, deposits, and bookkeeping.

4. Development of financial markets

- The first stock exchange was established in Amsterdam in the 17th century.
- Companies began issuing shares to raise capital.
- Financial markets expanded globally, enabling investment and economic growth

Modern Finance

Finance today includes banking, investments, insurance, and financial technology (fintech).

Global institutions, stock markets, and digital platforms dominate.

Concepts like corporate finance, public finance, and persona finance evolved.

INTRODUCTION TO FINANCE

Finance is an important field that contributes significantly to the functioning of moder economies. It includes a wide range of operations involving the management of money and other financial resources. Finance is a vital for making intelligent budgeting, investing, saving, and spending decisions at all levels, from individual homes to organizations and governments. At its foundation, finance is about knowing how money works and how to use it to achieve specific objectives. This includes raising cash in a variety of ways, such as issuing stocks on bonds, managing assets to achieve maximum return, and assessing risks to reduce potential losses.

1.2 DEFINITION OF FINANCE:

Finance is the science and practice of managing money and other financial assets. It entails the study, creation, and administration of funds, including bow money is raised, invested, and spent by individuals, businesses, and governments, Finance's fundamental purpose is to optimize resource allocation, control risks, and maximize investment value.

1.3 MEANING OF FINANCE:

Finance is the science and art of managing money and financial resources effectively, and the activity of earning, spending, saving, investing, and managing money.

1.4 SCOPE OF FINANCE:



- **Strategy and Planning** - A strategy is a comprehensive plan or method used to accomplish a certain goal or group of goals. It entails choosing how to compete in the market, where to put resources, and what steps to follow in order to accomplish long-term goals. Planning is the process of establishing objectives and deciding the actions required to accomplish them.
- **Treasury Management** - Treasury management is the oversight and control of an organization's financial resources, usually by its treasury department or function. The fundamental purpose of treasury management is to guarantee that the business has enough liquidity to meet its financial obligations and improve its overall financial performance.
- **Investor Relations** -Investor Relations (IR) is the strategic management department within a corporation that communicates with investors, analysts, shareholders, and other stakeholders. The major purpose of Investor Relations is to foster strong connections and open communication between the company and the financial community.

- **Investment Banking-** Investment banking is a type of banking that specializes in providing financial advice services, raising capital, and conducting mergers and acquisitions (M&A) for firms, governments, and other organizations.
- **Investment Analysis** - Investment analysis aims to help investors make informed decisions that optimize the risk-return trade-off while also aligning with their financial goals and preferences.
- **Investment Management-** Investment management, often known as asset management, is the professional management of various securities and assets (such as stocks, bonds, real estate, commodities, and other investments) on behalf of individuals or institutions.
- **Financial Planning** - Financial planning is a complete process that assists people, families, and organizations in assessing their financial condition, setting financial goals, and developing a strategy to accomplish those goals.
- **Real Estate** - The term "real estate" refers to property that includes land, buildings, natural resources, and other improvements attached or located on it. It is a physical asset class that includes a variety of residential, commercial, industrial, and agricultural properties.
- **Insurance-** Insurance is a financial agreement that protects against the possibility of financial loss. The risk of potential losses is transferred from an individual or corporation (the insured) to an insurance firm (the insurer) in exchange for the payment of a premium.
- **Commercial Banking** - Commercial banking encompasses the banking activities and services offered by financial institutions (banks) to enterprises, corporations, and government agencies.

1.5 ADVANTAGES OF FINANCE:

a) Ensures Availability of Funds

Finance helps in raising and allocating capital as per the needs of the business or individual ensures liquidity and smooth day-to-day operations.

b) Supports Business Growth and Expansion

Adequate financing enables companies to invest in new projects, expand operations, or enter new markets. Helps in technology upgrades, research and development, and hiring skilled employees.

c) Aids in Decision-Making

Financial analysis helps in making informed decisions about investments, expenses, pricing and profitability, Tools like budgeting, forecasting, and cost-benefit analysis are finance based.

d) Increases Profitability

Effective financial management ensures optimal use of resources, minimizes costs, and enhances profits.

e) Risk Management

Finance involves tools and strategies (like insurance, diversification, hedging) that help in identifying, assessing, and minimizing risks.

f) Enhances Investor Confidence

Sound financial planning and transparent reporting boost investor trust and attract more investment Improves company's reputation and market valuation.

g) Facilitates Long-Term Planning

Finance helps individuals and organizations plan for the future—like retirement, expansion, or sustainability

h) Improves Operational Efficiency

With proper budgeting and fund allocation, businesses can reduce waste, control spending, and streamline operations.

i) Promotes Economic Development

At a national level, finance supports infrastructure development, industrialization, and social programs, driving economic growth.

1.6 DISADVANTAGES OF FINANCE:

1. Risk of Over-Borrowing

Excessive reliance on debt can lead to high interest costs and financial distress.

2. Cost of Financing Raising finance

whether through loans, equity, or issuing bonds—often involves high costs such as:

- Interest payments
- Administrative or underwriting fees
- Dividend obligations

3. Mismanagement or Misallocation of Funds

Improper financial planning can lead to wastage, unproductive investments, or cash flow problems.

4. Dependency on External Sources

Relying too much on external finance can make a business or individual vulnerable to creditors or investors.

5. Interest Rate Fluctuations

Loans and credit are sensitive to interest rate changes. Rising rates increase the cost of borrowing, reducing profitability.

6. Market Volatility and Investment Risks

Financial markets are unpredictable. Investments can lead to losses due to market crashes, inflation, or poor economic conditions.

7. Complexity and Regulation

Finance involves complex calculations, compliance, and legal regulations that require expertise.

8. Short-Term Focus

Companies may focus on short-term financial gains (like boosting EPS) rather than long-term sustainability, leading to unstable growth.

1.7 TYPES OF FINANCE

- ❖ PERSONAL FINANCE
- ❖ CORPORATE FINANCE
- ❖ PUBLIC FINANCE

- ❖ **Personal Finance** - Personal finance refers to managing an individual's or household's financial activities, such as budgeting, saving, investing, insurance, and retirement planning
- ❖ **Corporate Finance**-Corporate finance refers to a company's financial actions and decisions involving capital management, funding sources, investments, and financial strategies in order to maximize shareholder value and meet organizational goals.
- ❖ **Public Finance** - Public finance is the study of government spending, revenue generation, and debt management at all levels of government (national, state, and local) and public entities. It includes the policies, methods, and mechanisms by which governments collect revenue, allocate resources, and manage their finances in order to achieve economic and social goals.
- ❖ **Investment Finance** -Investment finance is the branch of finance that focuses on allocating financial resources to various investment possibilities with the goal of creating profits for investors. It includes examining investment possibilities, managing portfolios, and making strategic decisions to maximize investment performance while mitigating risk.
- ❖ **International Finance** - International finance is the branch of finance concerned with monetary interactions between countries, such as foreign investments and exchange rates It includes international financial management, investment, and trade activities.

1.8 ORIGIN OF LEVERAGE:

- In the 20th century, the term began to be used figuratively, particularly in finance, to refer to using borrowed money (debt) to increase the potential return on investment
- Financial leverage thus parallels the mechanical idea: using a small input (your own capital) to achieve a greater effect (larger returns) by adding external resources (borrowed funds).

1.9 DEFINITION OF LEVERAGE:

Leverage refers to the strategic use of fixed costs or borrowed funds to increase the potential return on investment. It magnifies both gains and losses, depending on how effectively the underlying resources are managed.

1.10 MEANING OF LEVERAGE:

Leverage is the strategic use of borrowed money (debt) to increase the potential return on investment, acquire assets, or fund business operations. It enables individuals and businesses to amplify their financial capacity and pursue growth opportunities beyond what their existing capital would allow. When an entity is described as highly leveraged, its debt is more significant than its equity, which gives an idea of its financial structure. Leverage can be a very effective tool but also poses substantial risks.

1.11 ADVANTAGES OF LEVERAGE

i. Increased Returns on Equity

- Leverage can amplify profits when sales and operations perform well.
- Especially true for financial leverage—using debt can boost Earnings Per Share (EPS) without issuing more equity.

ii. Tax Benefits

- Interest on debt is tax-deductible, which reduces taxable income.
- This makes debt a cheaper financing option compared to equity.

iii. Retain Ownership

- Using debt (financial leverage) allows a business to raise funds without diluting ownership.
- 4. Operational Efficiency
- High operating leverage encourages efficient use of fixed resources.
- Once fixed costs are covered, additional sales contribute significantly to profit.

iv. Strategic Growth

- Leverage enables companies to expand faster by funding projects or acquisitions with borrowed capital.

1.12 DISADVANTAGES OF LEVERAGE

1) Increased Financial Risk

- Higher leverage means higher fixed obligations (like interest payments).
- This increases the risk of bankruptcy if revenue falls.

2) Volatile Earnings

- Small changes in sales can cause large fluctuations in net income due to leverage.
- Makes earnings less predictable and riskier.

3) Debt Covenants & Restrictions

- Borrowing often comes with contractual limits imposed by lenders.
- May restrict future borrowing, dividends, or strategic

4) Pressure During Downturns

- In tough economic times, companies with high leverage may struggle to meet obligations, risking default or forced asset sales.

5) Reduced Flexibility

- High fixed costs (operating leverage) limit a company's ability to cut expenses during downturns.

1.13 TYPES OF LEVERAGE



1.14 OPERATING LEVERAGE

DEFINITION

Operating leverage shows the sensitivity of a company's operating income to changes in sales. The more fixed costs a company has relative to variable costs, the higher its operating leverage.

Operating leverage refers to the degree to which a company uses fixed costs in its cost structure. It measures how a change in sales volume affects operating income (EBIT).

Formula:

$$\text{DOL (degree of operating leverage)} = \frac{\% \text{ Change in EBIT}}{\% \text{ Change in Sales Or at a specific level of sales}}$$

$$\text{DOL} = \frac{\text{Contribution Margin}}{\text{Operating income (EBIT)}}$$

Operating Income (EBIT) Where:

- Contribution Margin = Sales – Variable Costs
- EBIT = Earnings Before Interest and Taxes

KEY CONCEPTS

➤ High Operating Leverage:

- High fixed costs, low variable costs.
- Small increases in sales lead to large increases in operating income.
- Riskier, but potentially more profitable when sales grow.
- Example industries: airlines, software, manufacturing.

➤ Low Operating Leverage:

- Low fixed costs, high variable costs.
- More stable profits, but less benefit from sales growth.
- Example industries: consulting, retail, food service.

FEATURES OF OPERATING LEVERAGE:

I. Involves Fixed Operating Costs

Operating leverage arises from the presence of fixed operating costs (like rent, salaries, and depreciation) in the cost structure.

II. No Involvement of Financial Costs

like financial leverage, it does not involve debt or interest payments—it focuses purely on business operations.

III. Magnifies Effect of Sales on EBIT

A small change in sales can lead to a larger change in operating income (EBIT) due to fixed costs.

IV. Measured by Degree of Operating Leverage (DOL)

$$DOL = \% \text{ Change in EBIT} / \% \text{ Change in Sales}$$
It quantifies how sensitive EBIT is to changes in revenue.

V. High Risk, High Reward

High operating leverage means higher business risk, but also higher potential returns when sales are growing.

VI. Break-Even Point is Higher

Companies with high operating leverage have a higher break-even point, as more sales are needed to cover fixed costs.

ADVANTAGES OF OPERATING LEVERAGE

- **Higher Profit Margins After Breakeven**

Once fixed costs are covered, additional sales directly boost profit, since variable costs are low.

- **Cost Efficiency**

Encourages the use of automation, technology, and efficient systems, reducing the need for labour - intensive variable costs.

- **Scalability**

Ideal for scaling businesses—revenues can grow faster than costs, increasing profit significantly.

- **Predictable Cost Structure**

Fixed costs make it easier to forecast and budget, improving operational planning.

DISADVANTAGES OF OPERATING LEVERAGE

- **High Breakeven Point**

A company needs to generate higher sales to cover fixed costs, increasing business risk.

- **Earnings Volatility**

A small drop in sales can cause a large decline in operating income (EBIT), due to fixed costs not decreasing.

- **Reduced Flexibility**

Fixed costs are hard to cut in tough times (e.g., rent, salaries, depreciation), limiting adaptability.

- **High Risk in Uncertain Markets**

In industries with unpredictable demand, high operating leverage can lead to significant losses.

1.15 FINANCIAL LEVERAGE:

DEFINITION

Financial leverage measures the degree to which a firm uses debt in its capital structure. It amplifies the potential return on equity, but also increases the risk if earnings do not cover interest payments.

Financial leverage refers to the use of borrowed funds (debt) to finance the acquisition of assets or operations, with the goal of increasing returns to equity shareholders.

FORMULA:

$$\text{Degree of Financial Leverage (DFL)} = \frac{\% \text{ Change in EPS}}{\% \text{ Change in EBI}}$$

Or at a specific EBIT level:

$$\text{DFL} = \frac{\text{EBIT}}{\text{EBIT} - \text{Interest Expense}}$$

KEY CONCEPTS

➤ **High Financial Leverage:**

- A small change in EBIT results in a larger change in EPS (Earnings Per Share).
- Increases potential returns but also financial risk.
- Common in capital-intensive industries (e.g., real estate, utilities).

➤ **Low Financial Leverage:**

- Lower debt obligations.
- More stable earnings, lower risk.
- Common in service-based or tech companies that rely more on equity.

FEATURES OF FINANCIAL LEVERAGE

1. **Use of Debt Capital**

Financial leverage involves the use of debt in the capital structure, as opposed to only relying on equity.

2. **Fixed Financial Cost**

Debt financing introduces fixed interest payments, which must be paid regardless of the company's earnings.

3. **Enhances Return on Equity (ROE)**

When the return on investment (ROI) exceeds the cost of debt, financial leverage can magnify returns for equity holders.

4. **Increases Financial Risk**

Higher leverage increases the risk of default and bankruptcy during downturns or periods of low earnings.

5. **Impact on Earnings Per Share (EPS)**

Financial leverage affects the EPS of a company. Positive leverage increases EPS, while negative leverage decreases it.

6. Optimal Capital Structure

Leverage plays a key role in determining the optimal mix of debt and equity to maximize firm value.

7. Degree of Leverage

Companies can measure leverage through metrics like the Degree of Financial Leverage (DFL) to assess its impact on earnings.

8. Irrelevance in Perfect Markets (Theoretically)

According to Modigliani and Miller's Proposition I, in a perfect market with no taxes or bankruptcy costs, financial leverage is irrelevant to firm value.

ADVANTAGES OF FINANCIAL LEVERAGE

- **Increased Return on Equity (ROE)**

When returns from investments exceed the cost of debt, financial leverage amplifies profits for shareholders.

- **Tax Benefits**

Interest payments on debt are tax-deductible, reducing the company's taxable income and overall tax burden.

- **Preserves Ownership**

Raising funds through debt avoids issuing new shares, so existing shareholders retain control of the company.

- **Lower Cost of Capital**

Debt is usually cheaper than equity, especially in low-interest-rate environments, helping reduce the company's weighted average cost of capital (WACC).

- **Access to Additional Capital**

Enables companies to undertake larger projects or expand faster than they could using only internal funds or equity.

DISADVANTAGES OF FINANCIAL LEVERAGE

1. Increased Financial Risk

Companies must repay debt with interest, regardless of business performance. This raises the risk of default or bankruptcy.

2. Earnings Volatility

While leverage can amplify profits, it can also magnify losses, making earnings per share (EPS) more volatile.

3. Debt Covenants and Restrictions

Lenders often impose covenants, which may limit business decisions (e.g., limits on further borrowing or dividends).

4. Lower Credit Rating

High debt levels can lead to downgrades in credit ratings, increasing the cost of future borrowing.

5. Reinvestment Risk

If the borrowed funds do not generate expected returns, the company is still obligated to repay, putting strain on cash flow.

1.16 COMBINED LEVERAGE:

DEFINITION

Combined leverage reflects how both:

Operating leverage (due to fixed operational costs), and Financial leverage (due to interest on debt) together influence the company's bottom line (EPS).

Combined leverage (also known as total leverage) measures the total risk a company faces by combining the effects of operating leverage and financial leverage. It shows how a change in sales will affect the company's earnings per share (EPS).

FORMULA:

Degree of Combined Leverage (DCL) = DOL×DFL Or:

$$DCL = \frac{\% \text{ Change in EPS}}{\% \text{ Change in Sales}}$$

At a specific level of operations:

$$DCL = \frac{\text{Contribution Margin}}{\text{EBIT} - \text{Interest Expense}}$$

KEY CONCEPTS:

➤ **High Combined Leverage:**

- A small change in sales leads to a large change in EPS.
- Indicates high business + financial risk.
- Suitable for companies confident in stable and growing revenue.

➤ **Low Combined Leverage:**

- More stable earnings.
- Lower potential return, but lower risk.

FEATURES OF COMBINED LEVERAGE:

1. Combination of Operating and Financial Leverage

It reflects the cumulative effect of fixed operating costs (like rent, salaries) and fixed financial costs (like interest on debt).

2. Measures Total Business Risk

Combined leverage helps assess total risk—both operating (business) risk and financial risk.

3. Indicates Sensitivity of EPS to Sales Changes

It shows how a small change in sales can lead to a much larger change in EPS (Earnings Per Share).

4. Measured by Degree of Combined Leverage (DCL)

$DCL = DOL \times DFL$ This formula multiplies the Degree of Operating Leverage and Degree of Financial Leverage to find total sensitivity.

5. Useful in Decision Making

Helps managers and investors evaluate the impact of sales fluctuations on profits and risk, guiding capital and operational decisions.

6. High Combined Leverage Means High Risk

7. Firms with high combined leverage face greater variability in EPS, making them riskier, especially during economic downturns.

8. Relevant for Capital-Intensive Firms with Debt

Companies with high fixed costs and significant debt are more likely to exhibit high combined leverage.

ADVANTAGES OF COMBINED LEVERAGE

1. High Profit Potential

When both sales and operations perform well, combined leverage can significantly boost Earnings Per Share (EPS) due to the double effect of fixed operating and financial costs.

2. Strategic Use of Resources

Companies can optimize their capital and cost structures, using a mix of fixed costs and debt to maximize profitability.

3. Accelerated Growth

Combined leverage allows for faster expansion by financing growth with debt while utilizing fixed costs for efficiency.

4. Tax Efficiency

Interest expenses reduce taxable income, offering tax benefits that improve net income.

5. Competitive Advantage

Efficiently leveraged firms can operate with lower per-unit costs and higher returns, potentially outperforming less-leveraged competitors.

DISADVANTAGES OF COMBINED LEVERAGE

1. High Risk Exposure

Combined leverage amplifies total risk—if sales decline, both fixed operating costs and interest obligations remain, leading to sharp drops in EPS.

2. Greater Earnings Volatility

Even minor changes in sales can cause large swings in net income, making earnings less predictable.

3. Reduced Financial Flexibility

Fixed commitments from both operating and financial leverage can limit a firm's ability to adapt to downturns or changing market conditions.

4. Higher Breakeven Sales Volume

A high level of fixed costs means the company must generate more sales just to break even, increasing vulnerability in slow markets.

5. Risk of Overleveraging

Mismanagement or overly aggressive use of leverage can lead to insolvency, especially during economic downturns or cash flow shortages.

1.17 EARNINGS PER SHARE:



ORIGIN OF EPS

➤ Early 20th Century – Birth of Corporate Reporting

- As stock markets expanded in the early 1900s, especially in the U.S. and U.K., there was a growing need for standardized performance metrics to help investors evaluate company profitability.
- Companies began reporting net income, and financial analysts started dividing this by the number of shares to estimate earnings per share, although it was not standardized.

➤ 1930s – EPS Gains Prominence in the U.S.

- The Securities Exchange Act of 1934 in the United States led to mandatory financial disclosures by publicly traded companies.
- Analysts and investors increasingly used EPS as a key profitability measure, even before it was formally defined.

➤ 1970s – Formal Standardization

- The Financial Accounting Standards Board (FASB) in the U.S. issued standards (notably FASB Statement No. 128) to standardize the calculation of EPS, distinguishing between:

✚ Basic EPS

✚ Diluted EPS

- Similar standards were adopted globally by other accounting bodies.

➤ Modern Accounting – International Standards

- International Accounting Standard (IAS) 33 – Earnings Per Share governs EPS reporting globally.
- It requires consistent presentation and calculation of both basic and diluted EPS in financial statements of listed companies.

1.18 DEFINITION OF EPS:

Earnings Per Share (EPS) is a financial ratio that indicates how much net profit is attributable to each outstanding share of a company's common stock.

1.19 MEANING OF EPS:

Earnings Per Share (EPS) represents the portion of a company's profit that is allocated to each individual share of common stock. It gives shareholders an idea of how much money the company is making per share they own.

1.19 TYPES OF EPS:

1. Basic EPS

- Definition: It is calculated using net income available to common shareholders divided by the weighted average number of common shares outstanding during the period.

Formula:

$$\text{Basic EPS} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Weighted Average Common Shares Outstanding}}$$

- Use: Most commonly reported in financial statements.

2. Diluted EPS

- Definition: It accounts for all potentially dilutive securities (like stock options, convertible bonds, etc.) that could increase the number of shares and reduce EPS.

Formula:

$$\text{Diluted EPS} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Weighted Average Shares} + \text{Dilutive Securities}}$$

- Use: Gives a more conservative estimate of earnings per share if all possible shares were issued.

1.20 ADVANTAGES OF EPS

1. Simple Indicator of Profitability

EPS offers a clear, per-share view of a company's profitability, making it easy to understand for investors.

2. Widely Used Performance Metric

It is a standard measure used by analysts, investors, and financial media to compare company performance.

3. Helps in Investment Decisions

EPS is a core input for valuation ratios like the Price-to-Earnings (P/E) ratio, aiding in stock selection and valuation.

4. Tracks Earnings Growth Over Time

EPS trends can indicate whether a company's earnings are growing, stable, or declining, helping assess long-term performance.

5. Useful for Comparing Companies

Especially within the same industry, EPS helps in benchmarking financial performance.

6. Linked to Shareholder Value

A rising EPS often correlates with increasing shareholder returns and may result in higher dividends or share prices.

1.21 DISADVANTAGES OF EPS

1. Affected by Accounting Policies

EPS can be manipulated through accounting practices like revenue recognition, depreciation methods, or asset revaluation.

2. Ignores Capital Structure Differences

EPS doesn't reflect how much debt or equity a company uses, which affects financial risk.

3. Not Comparable Across All Companies

Different share counts, preferred dividends, or business models can make EPS less useful for comparing firms across sectors.

4. Excludes Cash Flow Information

EPS is based on net income, not cash flow, so it may overstate earnings if profits are not backed by real cash.

5. Distorted by Share Buybacks

Companies can boost EPS artificially by repurchasing shares, reducing the denominator without improving actual performance

CHAPTER-2

REVIEW OF LITERATURE

2.1 INTRODUCTION

Research design is the overall framework used to collect, measure, and analyze data. For this study, the research design focuses on examining how financial leverage influences Earnings Per Share (EPS) using quantitative financial data. A research design is the strategic plan that outlines the procedures for gathering and analysing data in a way that balances relevance to the research objectives with efficiency in execution. Essentially, the design framework of a research endeavour is commonly referred to as the research design. Various decisions such as what, where, when, and how to conduct a research study contribute to shaping the research design.

2.2 TITLE OF THE STUDY

**“A STUDY ON FINANCIAL LEVERAGE AND ITS IMPACT ON EPS
WITH REFERENCE TO MANDОВI MOTARS”**

2.3 STATEMENT OF THE PROBLEM:

Financial leverage, or the use of debt in a company’s capital structure, plays a crucial role in determining a firm’s profitability and financial risk. It can significantly impact Earnings Per Share (EPS), a key indicator for investors. This study focuses on analysing the relationship between financial leverage and EPS with specific reference to Mandovi motors. Despite its importance, limited research exists on how leverage affects EPS in the automobile dealership sector. Understanding this link is essential for making informed financing decisions. The study aims to assess whether higher leverage improves or reduces EPS and its overall impact on the company’s financial performance.

2.4 OBJECTIVES OF THE STUDY:

- ✚ To study the Effect of degree of financial leverage on the return on investment.
- ✚ To Study the impact of financial leverage on R.O.A & R.O.E and Earnings.
- ✚ To Estimate the effect of changes in earnings due to financial leverage of the company.
- ✚ To analyse EPS over a period of time.

2.5 SCOPE OF THE STUDY:

The scope of this study is to evaluate the relationship between financial leverage and its effect on Earnings Per Share (EPS) and the data is drawn to analyse is from the audited reports of Mandovi motors in the branch of Gowribidanur.

2.6 REVIEW OF LITERATURE:

(John Davies Odhiambo, 2025)¹

The literature review in this document centres around two foundational theories: the trade off theory and the pecking order theory. It discusses how firms must balance debt usage to avoid distress (Farooq et al., 2023) while recognizing that internal funds are typically preferred (Brealey et al., 2018). Empirical insights by Sugumar (2023) identify firm size, profitability, and dividend payout as primary leverage determinants. Afandy (2024) supports that leverage and cost management significantly influence performance in Indonesia's hospitality sector. The document stresses the risk-return balance and highlights sectoral variability in leverage effects, suggesting tailored financial strategies based on firm and industry characteristics.

(Rizqa Anita, 2023)2

The 2023 study explores the impact of leverage on firm value, particularly during the COVID-19 pandemic. The literature reviewed highlights the roles of return on equity (ROE) and earnings per share (EPS) as performance indicators. Prior studies show mixed findings—some firms benefited from increased leverage during the pandemic, while others did not. Shen et al. (2020) noted a downturn in high-risk sectors like tourism. The Jakarta Composite Index data illustrates market recovery in 2021. The review underscores that leverage indirectly affects firm value, primarily through mediating performance metrics like ROE.

(Radmehr, 2023)3

This comprehensive study evaluates how various leverage metrics (equity multiplier, interest coverage, capitalization ratio, etc.) affect the financial performance of Japanese nonfinancial firms. It uses the generalized method of moments on data from 263 firms over two decades. Results show that some leverage proxies like equity multiplier and interest coverage positively affect ROA, ROE, and EPS, while others such as degree of financial leverage and capitalization ratio negatively impact them. The study supports trade-off theory and suggests that while debt can enhance performance, excessive reliance increases financial risk.

(Sapiri, 2023)4

This paper investigates the effect of operating and financial leverage on EPS in Indonesia's real estate sector. The findings indicate that while operating leverage significantly improves EPS, financial leverage alone does not show a significant effect. However, when combined, both types of leverage positively influence EPS. This implies that firms must carefully coordinate their capital and operational strategies to enhance profitability. The study advises investors

to consider EPS as a key metric when evaluating firms and highlights the importance of prudent debt usage in capital planning.

(Nafia Laila, 2021)⁵

This paper provides a detailed theoretical framework on the types and effects of leverage— operating, financial, and combined. Drawing on classical theories like Modigliani & Miller (1958, 1963) and Myers (1984), the study explains how high leverage magnifies both risk and return. Mandelker & Rhee (1984) noted that combined leverage could lead to significant EPS fluctuations. The literature also includes empirical observations, such as Gleason et al.'s finding of a negative relationship between financial leverage and profitability in European firms. The review supports a nuanced view of leverage management as a double-edged sword.

(Gentry, 2021)⁶

Gahlon and Gentry's seminal work explores how systematic risk (beta) is influenced by both operating leverage (DOL) and financial leverage (DFL). The authors argue that the capital asset pricing model (CAPM) alone is insufficient to explain beta without considering the real economic decisions of firms. They integrate DOL and DFL into a valuation framework to demonstrate how variations in fixed costs and financing structures affect systematic risk. The study proposes that higher DOL and DFL amplify the sensitivity of a firm's cash flow to market conditions, thus increasing beta. Unlike prior models that only use historical returns, this approach ties risk directly to real-asset variables.

(Chen, 2020)⁷

This paper provides a thorough literature review focusing on the relationship between financial leverage and firm performance, particularly in Chinese listed companies. Earlier studies, such as Hamada (1972), emphasized that

higher leverage correlates with increased systematic risk. Conversely, newer research explores how leverage affects returns. Ozdagli (2012) and Obreja (2013) argue that higher leverage reduces stock return volatility, with operating leverage moderating this relationship. Firm performance determinants, such as human resource quality and CSR (Calantone et al., Wright et al., Mackey et al.), also feature prominently. Despite mixed findings—some showing a negative, others a positive correlation between leverage and performance—the consensus leans toward a nuanced, context-dependent interpretation.

(Dr. R. Kabir, 2020)8

This paper reviews determinants of capital structure and their varying effects across firms and countries. It highlights firm-specific attributes such as size, profitability, liquidity, and asset structure, noting their inconsistent influence on leverage. Studies like those by De Jong et al. (2008) and Ramli et al. (2019) show cross-country variability. Liquidity and profitability generally reduce leverage, while size often increases it. Empirical evidence from the automotive industry during financial crises reveals that firms must reassess capital structures under economic stress. The review concludes that there is no one-size-fits-all leverage model due to fluctuating macroeconomic conditions and firm-level differences.

(Nilam Rana, 2019)9

This study reviews leverage analysis with a focus on Hero MotoCorp, identifying operating and financial leverage as key tools for evaluating profitability. The literature emphasizes how financial statement analysis can uncover a firm's strengths and weaknesses. Operating leverage is shown to affect returns by managing assets without ownership. The review also highlights the importance of understanding financial statement data over time to make strategic financial decisions. It indicates that leverage can improve

returns if managed properly, though it notes data limitations and lack of inter-firm comparisons.

(Dr. Joji Abey, 2019)¹⁰

This literature review assesses how capital structure decisions impact financial leverage in developing countries. It includes foundational works like Booth et al. (2001), emphasizing size, asset structure, and profitability as critical factors. Other studies, such as Bhaduri (2002) and Mohanty (2003), focus on the roles of growth, cash flow, and R&D expenditures in determining leverage. Voulgaris and Asteriou (2004) find that large firms benefit from efficient asset use, whereas SMEs are more influenced by liquidity and fixed assets. Overall, the review portrays leverage as context-dependent, shaped by firm characteristics and market conditions.

(Trie Nadilla, 2019)¹¹

This study focuses on the impact of financial leverage and earnings per share (EPS) on earnings management in Indonesian firms listed on the LQ45 index. It uses discretionary accruals to measure earnings management and applies panel data regression over three years. Findings indicate that leverage negatively affects earnings management, implying that higher debt may constrain manipulative practices. Conversely, EPS positively influences earnings management, suggesting that firms may inflate earnings to meet shareholder expectations. The research underscores the need for stringent financial controls and transparency to curb earnings manipulation in high-performing firms.

(Berlian Karlina, 2019)¹²

The study examines how financial leverage, firm size, basic earning power, and activity ratio impact earnings per share (EPS) in Indonesian automotive firms from 2013–2017. It concludes that financial leverage, firm size, and basic

earning power significantly influence EPS, while activity ratio does not. The findings support the notion that leverage and internal efficiency metrics play a crucial role in determining profitability. The research applies multiple regression analysis and suggests that maintaining an optimal debt-equity mix and improving operational performance can enhance shareholder value.

(Francesca Magli, 2018)¹³

This paper evaluates the impact of the new lease accounting standard IFRS 16 on financial leverage and firm performance. The research shows that recognizing lease liabilities and right-of-use assets on the balance sheet leads to an increase in financial liabilities and EBITDA, while equity tends to decrease. These changes significantly affect key financial ratios like debt-to-equity and debt-to-total assets. The paper warns that sectors reliant on operating leases, such as airlines and retail, will see the most drastic changes. It emphasizes that IFRS 16 improves transparency and comparability but also calls for firms to reconsider lease-versus-buy decisions and adjust communication strategies accordingly.

(Sarkar, 2018)¹⁴

Sarkar's theoretical study critically revisits the traditional view that operating and financial leverage are substitutes. Using a contingent-claim framework, the paper finds that the relationship between the two is far more complex and can be positive, negative, or even neutral depending on which variable (DOL or DFL) is exogenously fixed. It introduces a model where both leverage types are endogenously determined and highlights that managerial decisions around capacity and capital structure are intertwined with risk profiles. The paper challenges conventional wisdom and provides clarity on conflicting empirical findings by showing that leverage relationships depend heavily on the firm's strategic context and cost structures.

(Prabhavathi, 2016)15

This empirical research evaluates how various forms of leverage—operating, financial, and combined—affect shareholders' wealth in Indian automobile firms. Utilizing panel data from 12 NSE-listed companies from 2003–04 to 2012–13, the study applies both simple and multiple regression techniques. Findings reveal that financial leverage significantly influences metrics such as Return on Capital Employed (ROCE), Return on Equity (ROE), and Net Worth, indicating that debt plays a crucial role in determining shareholder returns. The paper also integrates prior international and domestic studies to contextualize leverage decisions. It highlights that while leverage can boost returns, excessive use may amplify risk, reinforcing the need for balanced financial planning.

(Seraina C. Anagnostopoulou, 2016)16

This study contributes to the literature by exploring how financial leverage influences a firm's choice between real earnings management (RM) and accrual-based earnings management (AM). Prior research suggests a substitution effect between RM and AM, with firms opting for the method that is less detectable or costly under scrutiny. The paper finds that higher leverage leads firms to prefer RM over AM due to RM's lower visibility to external parties like investors or regulators. However, in highly leveraged firms, both forms of earnings management may coexist, indicating a complementary relationship.

(Hiran, 2016)17

This paper investigates the relationship between liquidity, leverage, and profitability in 25 Indian automobile companies over 2011–2015. The analysis, supported by SPSS-based statistical tests, reveals that liquidity indicators (current and quick ratios) have varied effects on profitability, with inventory turnover showing a negative association with operating profit. On leverage,

both operating and financial leverage are found to negatively influence profitability, while combined leverage shows no significant impact. The findings imply that excess fixed obligations can deteriorate performance, especially in volatile markets. The study calls for strategic balancing of liquidity and leverage to enhance overall financial health and shareholder returns.

(Sachdeva, 2014)¹⁸

This study explores the influence of profitability and financial leverage on the capital structure of Indian automobile companies over a ten-year period (2009–2018). By analysing data from 10 listed firms using regression and correlation techniques, the paper finds that profitability has a low negative relationship with capital structure, indicating that more profitable firms tend to use less debt. Conversely, financial leverage exhibits a mild positive correlation with capital structure. The study supports the pecking order theory, suggesting firms prefer internal financing over debt. It also emphasizes the strategic use of leverage and profitability indicators to optimize financial decision making. This empirical insight provides valuable implications for managers in balancing risk and return in capital structuring.

(Prabakaran, 2013)¹⁹

This early study investigates the relationship between financial leverage and earnings per share (EPS), using UCAL Polymer Industries as a case study. It critiques the general applicability of the leverage theorem, showing that UCAL achieved EPS growth through reverse leverage. Referencing foundational theories, the paper argues that firms must balance debt and equity carefully. The review includes insights from Pandey (1999) and Raj Dhankar (1996), who examined leverage dynamics in Indian firms. The study suggests that leverage outcomes are not universally predictable, and firm-specific strategies play a key role.

(Gill, 2012)²⁰

This paper explores the determinants of financial leverage among small business firms in India, focusing primarily on businesses in Punjab. The authors use survey-based research to identify key variables such as business growth, performance, total assets, sales, tax considerations, and family influence. Their findings show a positive relationship between these variables and the firms' use of debt, indicating that growing and asset-rich firms tend to leverage more. The study critiques traditional capital structure theories, suggesting that they often overlook the constraints and behaviors unique to small firms, such as limited access to equity markets and reliance on informal funding. It also highlights the influence of asymmetric information, agency costs, and collateral availability on financing decisions.

(Ajmera, 2011)²¹

This study examines the dynamic relationship among financial leverage, earnings, and dividends for Maruti Suzuki India Ltd. The authors conduct an empirical analysis using financial data to assess how leverage impacts firm profitability and dividend payouts. They argue that financial leverage can be a double-edged sword—enhancing returns during profitable periods while magnifying losses during downturns. Their analysis indicates a significant correlation between leverage and earnings, suggesting that firms with stable and significant correlation between leverage and earnings, suggesting that firms with stable and strong earnings are more inclined to take on debt. The paper also highlights how leverage decisions affect dividend distribution, implying that firms with higher debt might adopt conservative dividend policies to preserve liquidity.

(Rayan, 2008)²²

This research investigates whether financial leverage affects firm value in the South African context, referencing theories such as Modigliani-Miller's

theorem, the trade-off theory, and the pecking order theory. Using panel data from JSE-listed companies, it concludes that higher financial leverage correlates negatively with firm value. It also studies the influence of interest rate volatility on capital structure, though findings here are inconclusive. The study examines how firm characteristics like profitability, size, and asset structure interact with leverage decisions. It highlights that the effect of leverage is industry-dependent and suggests that capital structure optimization requires a balance of debt benefits and financial risk.

(Kumar, 2005)²³

This review critically investigates the factors affecting financial leverage through a comprehensive study of 107 papers published from 1991 to 2005. The paper highlights that although several theoretical frameworks like Modigliani-Miller's irrelevance theory, static trade-off theory, pecking order theory, and agency cost theory attempt to explain leverage decisions, no single universal determinant exists. The review categorizes literature under multiple frameworks, including legal environment, mean reversion (target leverage), and transaction cost theories, offering insights into complex influences on capital structure. It further notes the lack of consensus in existing research and calls for integrated models that combine these frameworks agency conflicts. Future research should focus on synthesizing these determinants into unified models and understanding leverage in networked environments.

(SARITAS, 2000)²⁴

This study explores the relationship between financial leverage and a firm's return on equity (ROE) and risk exposure. It emphasizes that financial leverage can boost ROE when return on assets (ROA) exceeds the cost of debt but can also amplify shareholder risk. The paper differentiates between business risk (from operations) and financial risk (from debt usage). It highlights that short-term debt is riskier than long-term debt due to limited risk-sharing

mechanisms. The research suggests that financial leverage has dual effects: enhancing profitability in good years and exacerbating losses in poor economic periods. Business risk factors such as demand volatility, price fluctuations, and input costs are also considered.

(Prezas, 1987)²⁵

Prezas' paper investigates the interrelationship between debt and the degrees of operating leverage (DOL) and financial leverage (DFL), challenging traditional views that treat these elements independently. Through a theoretical model, the study demonstrates that changes in a firm's capital structure can simultaneously influence both DOL and DFL, especially when production and financing decisions are not separable. The analysis reveals that capital structure affects input costs and, consequently, the production process. This dynamic interaction implies that increasing debt may not always result in increased DFL alone; it can also alter DOL depending on how input costs respond. The study argues that the firm's risk profile shifts holistically with leverage, meaning both operational and financial risks are endogenously linked. By incorporating debt elasticities of real capital and contribution margins, the paper presents a nuanced understanding of leverage impact, contributing to a more realistic framework for risk and capital structure analysis.

2.7 RESEARCH GAP:

- **No past study on 5-year financial leverage trends:** There is no existing research analysing how Mandovi motors leverage changed over the last 5 years.
- **Lack of 5-year EPS impact evaluation:** The relationship between financial leverage and EPS over a consistent 5-year period has not been studied for this firm.

- **Missing year-wise correlation analysis:** No study has investigated how changes in leverage have influenced EPS each year from 2021 to 2025.
- **Absence of dealership-focused time-series data:** Research on automobile dealers like Mandovi Motors rarely includes multi-year financial data for deep analysis.
- **Limited contextual insights:** The effect of leverage on EPS during events like market fluctuations or internal business changes over 5 years is underexplored.

2.8 OPERATIONAL DEFINITION OF CONCEPTS:

1. LEVERAGE:

The ability of a firm to use fixed cost assets or fund to magnify the returns is termed as leverage. The higher leverage obviously implies higher outside borrowings and hence it is riskier if the firm's earning capacity is reduced. In other words, only when the Return on Investment is higher than the cost of outside Financial Leverage and its Impact on Earning Per Share borrowing, the effect of leverage will be favourable. There are two types of leverage. (a) Operating Leverage. (b) Financial leverage.

2. OPERATING LEVERAGE:

It is one of the techniques to measure the impact of changes in sales which lead for change in the profits of the company. If any changes in the sale it will lead to corresponding changes in profit. It helps to identify the position of fixed cost and variable cost. The DOL is defined as the percentage change in the EBIT relative to a given percentage change in sales. EBIT depends on sales. A change in sales will affect EBIT. The variability in EBIT due to a change in sales is affected by the composition of fixed and variable costs.

$$\text{DOL} = \frac{\text{Percentage change in EBIT}}{\text{Percentage change in sales.}}$$

3. FINANCIAL LEVERAGE:

Financial leverage measures firm's exposure to the financial risk. The use of the fixed charges sources of funds, such as debt and preference capital along with the owners' equity in the capital structure, is described as financial leverage, gearing, or trading on equity. The financial leverage employed by a company is intended to earn more return on the fixed-charge funds than their costs. The surplus (or deficit) will increase (or decrease) the return on the owners' equity. The rate of return on the owners' equity is levered above or below the rate of return on total assets. The percentage change in EPS occurring due to a given percentage change in EBIT is referred to as the DFL.

$$\text{DFL} = \frac{\text{Percentage changes in EPS}}{\text{Percentage changes in EBIT.}}$$

4. COMBINE LEVERAGE:

Operating and financial leverages together cause wide fluctuation in EPS for a given change in sales. It can be done by multiplying the operating leverage and financial leverage. The operating leverage affects the EBIT and the financial leverage affects the EPS. The management has to devise a right combination of the operating and financial leverage. A company whose sales fluctuate widely and erratically should avoid use of high leverage since it will be exposed to a very high degree of risk.

5. EFFECT OF HIGH DEGREE OF LEVERAGE:

High degree of leverage helps the firms to increase the revenues in the short run and also helps in increasing the share holders' wealth in the long run. However, it is more risky for a company to have a high ratio of financial leverage as the commitment towards the payment of interest and other debt charges may create a financial crunch for the organization. This in-turn also affects the credit rating of the firm in the market.

6. EFFECT OF LOW DEGREE OF LEVERAGE:

It is less risky for a company to have a low ratio of financial leverage. With a low leverage company can meet its debt obligations and there is an opportunity for it to find new lenders in the future, however, the operating income may also be less with low degree of leverage. Another advantage is that the firm may have good credit rating in the market and hence can find the financiers easily.

7. FINANCIAL PERFORMANCE WITH LEVERAGE:

Financial performance simply can be defined as the performance of the organization in meeting its long term and short term financial objectives of the firm. The financial objectives are said to be met if the firm is able to comply with all financial needs of the short term and long term effectively without any friction or risk. It is the general wellbeing of the firm in terms of finances. The financial performance is measured through Return on Investment (ROI), Return of Equity (ROE) and Earnings per Share (EPS) etc

8. EARNINGS PER SHARE (EPS)

EPS represents the portion of a company's profit allocated to each outstanding equity share. For Mandovi motors, it is calculated using

the formula: $EPS = \frac{\text{Net Profit After Tax} - \text{Preferred Dividends}}{\text{Number of Equity Shares Outstanding}}$ EPS is used as an indicator of profitability and shareholder value in this study.

9. CAPITAL STRUCTURE:

Capital structure is defined as the mix of debt and equity used by Mandovi motors to finance its assets. It includes both short-term and long-term borrowings and shareholder equity. It is analysed to understand the degree of financial leverage.

10. CASHFLOW STATEMENTS

The analysis of cash flow statements in the automobile perspective is crucial for understanding a company's financial health and operational efficiency within this capital-intensive and cyclical industry. It goes beyond the income statement, which can be influenced by non-cash items, to reveal the actual cash inflows and outflows, providing insights into a company's liquidity, solvency, and ability to fund future growth.

11. PROFIT & LOSS A/C

The analysis of a Profit & Loss (P&L) account in the automobile perspective involves a deep dive into a car manufacturer's or automotive parts supplier's financial performance over a specific period. It goes beyond simply looking at the bottom line (net profit) and examines the various components that contribute to profitability, providing valuable insights for stakeholders.

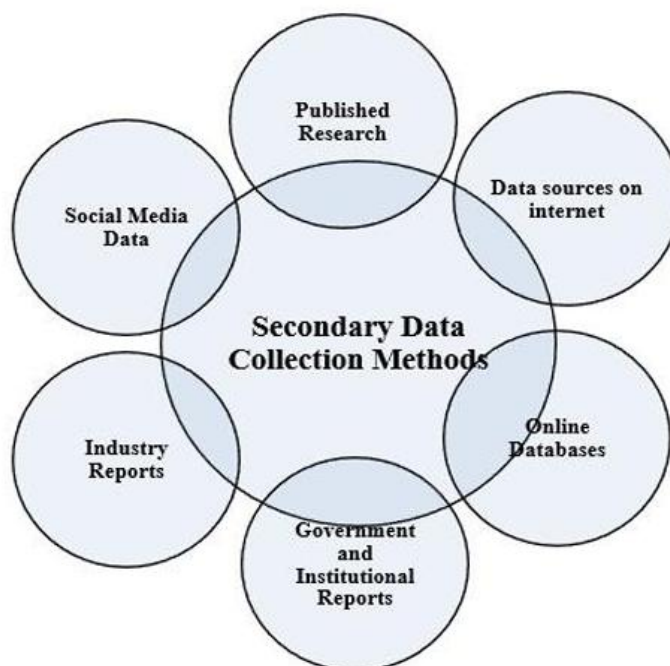
12. BALANCE SHEET

A balance sheet is a type of financial document that shows the state of a business's finances as of a particular date. It displays an organization's assets, liabilities, and equity- the difference between the two.

2.9 NEED FOR THE STUDY:

This study is essential to understand how financial leverage affects Earnings Per Share (EPS) at Mandovi motors. In today's competitive environment, efficient use of debt can enhance profitability but also increases financial risk. Analyzing this relationship helps the company make better capital structure decisions. The study provides insights for management and investors to evaluate the impact of leverage on financial performance. It supports strategic planning aimed at maximizing shareholder value.

2.10 SOURCES OF DATA COLLECTION:



SECONDARY DATA

Secondary sources, on the other hand, compile information from organizations or entities that have already collected it from primary sources. These sources analyse, explain, or combine primary data to provide a broader perspective. Examples of secondary sources are critiques, official surveys, scholarly publications, and academics texts.

2.11 SAMPLING:

This project is completely on Secondary data, so primary data is not been used to obtain the sample size.

2.12 SAMPLE DESIGN OF DATA COLLECTION TOOLS:

The tools and techniques utilized for data analysis involve interpreting data through various methods. These methods include:

- Interacting with Company's Finance Department and collecting financial data.
- Utilizing tables, analysis sheets, and bar graphs to identify solutions to problems.

2.13 RESEARCH METHODOLOGY:

1. Research Design:

The study adopts a quantitative and analytical research design, focusing on secondary financial data of Mandovi motors over a 5-year period (2021–2025). It aims to analyse the impact of financial leverage on EPS using various financial tools and techniques.

2. Nature of Study:

This is an analytical and descriptive study based on historical financial data to evaluate trends, relationships, and financial performance indicators.

3. Data Collection:

- Type of Data: Secondary Data
- Source of Data: Audited financial statements, profit and loss accounts, balance sheets, and internal reports of Mandovi motors for the years 2021 to 2025.

4. Tools and Techniques Used for Analysis:

➤ Accounting Calculations:

- Contribution
- EBIT (Earnings Before Interest and Taxes)
- EBT (Earnings Before Tax)
- EAT (Earnings After Tax)
- EPS (Earnings Per Share)
- Operating Leverage
- Financial Leverage
- Combined Leverage
- Return on Assets (ROA)
- Return on Investment (ROI)
- Return on Equity (ROE)

➤ Comparative Analysis:

- Year-wise comparative analysis of Contribution, EBIT, EBT, EAT, and EPS to understand performance trends over the five years.

➤ Consolidated Statement Preparation:

- Preparation of a consolidated statement incorporating all the calculated metrics (Contribution, EBIT, EBT, EAT, EPS, ROA, ROI, ROE, OL, FL, CL) for a holistic view.

➤ **Correlation Analysis:**

Correlation between:

- Operating Leverage and EPS
- Financial Leverage and EBIT, EBT, EAT, and EPS

This is used to evaluate the strength and direction of relationships between leverage and profitability indicators.

2.14 REFERENCE PERIOD:

The study is limited to data such as Cashflow statement, P&L A/c, Balance sheet, for the four years 2021-2025 of Mandovi motors.

2.15 Limitations Of The Study:

- **Limited to One Company:**
The study focuses only on Mandovi motors, so the findings may not be generalizable to other companies or sectors.
- **Depended on Secondary Data:**
The analysis is based on secondary data from financial reports, which may have limitations in accuracy or completeness.
- **Time Frame Constraints:**
The study covers financial data from a specific period, which may not reflect long-term trends or future performance.
- **Excludes Non-Financial Factors:**
Operational, managerial, and market-related factors affecting EPS are not considered in this study.

A STUDY ON FINANCIAL LEVERAGE AND ITS IMPACT ON EARNING PER

SHARE WITH REFERENCE TO MANDОВI MOTORS

